# Danh Tran

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## EDUCATION

**California State Polytechnic University, Pomona** Expected May 2029
Bachelor of Science in Mechanical Engineering, GPA: 3.81/4.00

## EXPERIENCE

**Robotics Engineer Research Intern** | Python, C++, MATLAB, ROS, Motion Planning, Linux December 2025 – Present
CREST Grant Robotics Research Lab Pomona, CA

- Saved ~$4,000 by sourcing missing robotic manufacturing parts, comparing in-house SolidWorks design and machining against market-ready parts, and negotiating vendor discounts
- Integrated a HaptX haptic glove with a Sarcomere Dynamics ARTUS Lite robotic hand through a cross-platform ROS bridge (Windows ↔ Linux, UART), enabling real-time bidirectional teleoperation with force feedback
- Building a ROS 2 Jazzy URDF model and joint controllers for an Epson T3-B SCARA robot in Gazebo, defining kinematics, joint limits, and command interfaces for a digital twin of a 4-station manufacturing cell
- Commissioned and programmed multiple EPSON robots running simultaneously in one cell with Siemens PLC ladder logic
- Authored equipment setup procedures and lesson plans for an undergraduate robotic manufacturing course; recognized as the lab's most valuable student

**Robotics Engineer Research Intern** | KiCAD, MATLAB, Autodesk Fusion, SolidWorks May 2026 – August 2026
CREST-MECIS Robotics Research Lab Edinburg, TX

- Co-designed a UAV with a modular 3D-printed frame, cutting frame weight ~50% through Fusion generative-design topology studies, FEA, and physical load testing of printed parts
- Validated a 15 lb payload capacity and 20 to 30 minute unloaded flight endurance for the UAV platform
- Designed hardware schematics and PCB layouts in KiCAD for a UAV-deployed structural crack detection sensor
- Improved crack detection accuracy to 1 mm with ~0.5 mm uncertainty by analyzing laser crack sensor data from multiple 3D-printed samples in MATLAB
- Machined 1215 steel mill and lathe parts to ±0.005 in. from drawings, including tapping and single-point threading

## PROJECTS

**Pickleball Launcher** | Raspberry Pi, ESP32, Python, C++, BLE, 3D Printing

- Scoped a pickleball training launcher with a practicing engineer, using Raspberry Pi AI cameras to detect players and adjust shot difficulty to skill level
- Cut ~$200 in electronics and lowered energy use by comparing 3D-printed gears against capstan drives to optimize the ball feed rate and increase durability
- Developed the physical architecture in SolidWorks to fit the full mechanism in a 40x40x80 cm envelope, ~50% smaller than most commercial pickleball machines

**Acoustic Attenuation Device** | SolidWorks, Thermodynamics, 3D Printing

- Designed a passive acoustic attenuation system for a pneumatic airsoft platform using wave interference
- Reduced acoustic signature ~50% by modeling compressible gas flow with adiabatic relations and ideal gas energy equations to redistribute peak pressure over time
- Cut post-processing time by ~5 minutes per unit by designing a support-free 3D-printable part with a 45-degree maximum overhang in ABS at 1.6 mm wall thickness

## SKILLS

**Electrical:** PCB Design, Altium, KiCAD, ESP32, Arduino, Raspberry Pi, CAN Bus, Motor Control, PWM, I2C, SPI, UART, BLE, RFID
**Software:** Python, C++, Bash, Java, JavaScript, HTML, CSS, React, Three.js, MATLAB, ROS2, Git, Excel, Linux, Gazebo, URDF
**Mechanical:** SolidWorks, Autodesk Fusion, Generative Design, FEA, 3D Printing, Manual Mill & Lathe, Threading/Tapping, GD&T & Tolerancing, Precision Measurement, Electromechanical Integration